

LLM Benchmarking: Context-Based QA Performance Analysis

Benchmarking six leading language models across 50 historical context-based questions, with performance measured through confidence scoring across three distinct question formats.

Executive Summary

79.5	77.1	71.9	62.4
Qwen3:1.7b	Llama3.2:latest	Gemma3:1b	Qwen2.5vl3b
Top performer with 82.6%, 79.1%, 76.8% confidence across three question sets	In second place with 77.9%, 80.8%, 72.5% scoring profile	Third position with balanced 71.2%, 75.3%, 69.1% scores	Even distribution at 62.4% across all question sets

Performance Ranking Overview

Naming Conventions

gpt-4o-mini 70.5%, 71.0%, 27.8% 56.4% average	gpt-5-mini 62.0%, 56.4%, 27.8% 48.7% average	qwen3:1.7b 82.6%, 79.1%, 76.8% 79.5% average	qwen2.5vl3b 62.4%, 62.4%, 62.4% 62.4% average
llama3.2:latest 77.9%, 80.8%, 72.5% 77.1% average			

Additional Insights

 **Visual reference only – replace with actual score data**

The following chart displays Qwen3:1.7b (79.5 points), Llama3.2:latest (77.1 points), Gemma3:1b (71.9 points), Qwen2.5vl3b (62.4 points), GPT-4o-mini (56.4 points), and GPT-5-mini (48.7 points). Actual score distribution across the three question sets may vary.

Benchmark Details

Test parameters: 50 historical context-based questions answered by 6 LLMs across 3 question sets with 3-way confidence scoring and averaging

 **Potential analysis opportunities**

Dataset reveals score distribution patterns and potential outliers across the question sets.